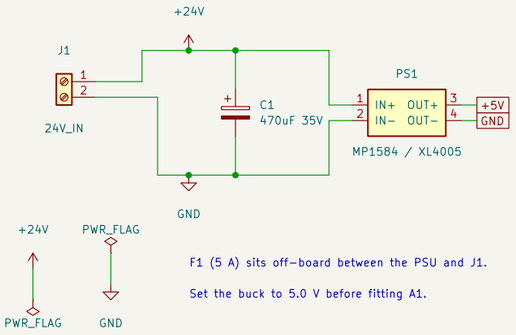
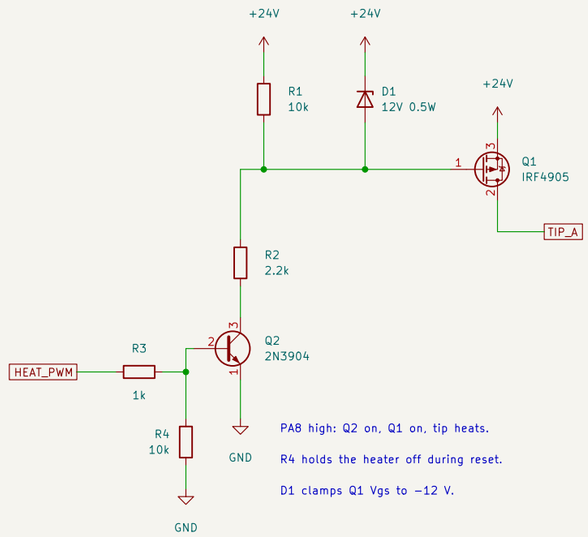
## Power input




F1 (5 A) sits off-board between the PSU and J1.

Set the buck to 5.0 V before fitting A1.

## Heater high-side switch

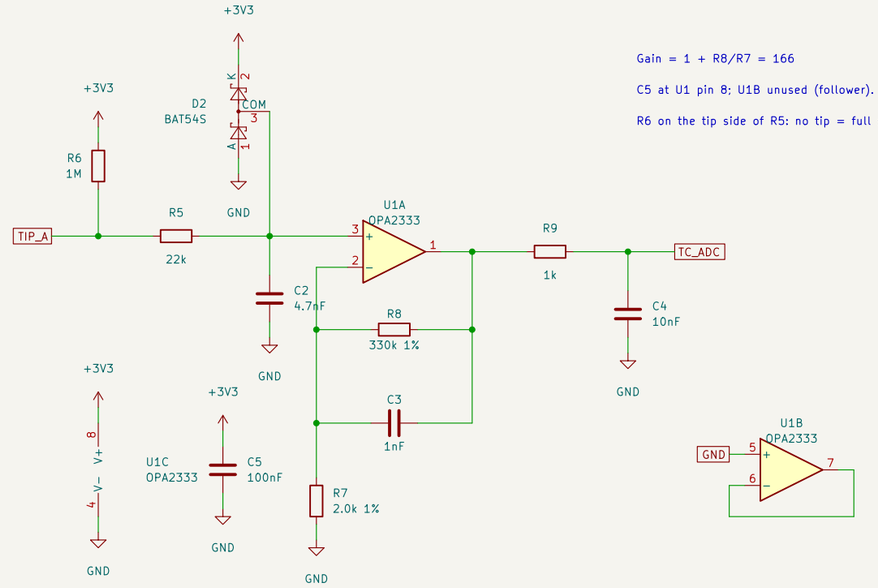



PA8 high: Q2 on, Q1 on, tip heats.

R4 holds the heater off during reset.

D1 clamps Q1 Vgs to −12 V.

## Thermocouple amplifier

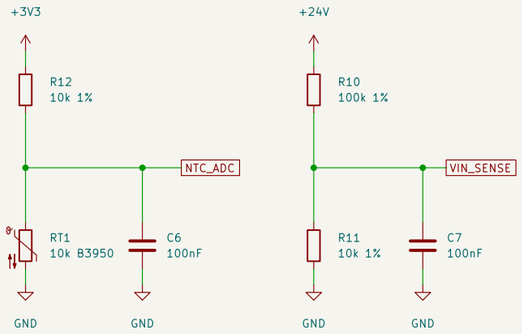



Gain = 1 + R8/R7 = 166

C5 at U1 pin 8; U1B unused (follower).

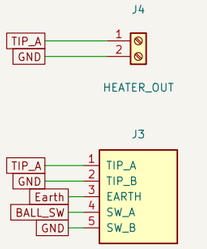
R6 on the tip side of R5: no tip = full scale.

## Sensing




RT1 is wired off-board, glued to the GX16 socket.

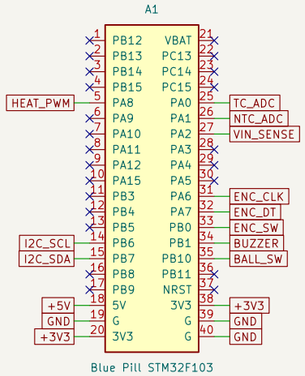
## Handle connection




J4 pin 2 (tip B) is the star ground point.

J3 pin numbers are placeholders: check your handle with a multimeter. EARTH goes to the PSU earth terminal and case, never to GND.

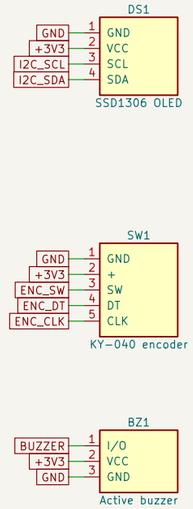
## MCU module




PA9/PA10 free for a serial debug log.

Pin order follows the common Blue Pill silkscreen.

## User interface




All GND returns meet at one star point: tip terminal B (J4 pin 2). Keep logic current off that wire.

Matches the T12 Station Build Guide and Perfboard BOM

STM32 Blue Pill + modules, perfboard analog stage

**Module-based build**

Sheet: /

File: T12_Station.kicad_sch

# Title: T12 Soldering Station

| Size: A3 | Date: 2026-10-28 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |